使用JARVIS、Cloud Sync VMware及Nemo建置虛擬助理

NetApp Solutions

NetApp
May 17, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/netapp-solutions/ai/cainvidia_jarvis_deployment.html on May 17, 2022. Always check docs.netapp.com for the latest.

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總覽

本節詳細說明虛擬零售助理的實作方式。

"下一步：JARVIS部署"

JARVIS部署

您可以註冊 "JARVIS早鳥方案" 以存取NVIDIA GPU Cloud（NGC）上的JARVIS容器。從NVIDIA接收認證之後、您可以使用下列步驟來部署JARVIS：

1. 登入NGC。
2. 將您的組織設在NGC：「ea - 2-jarvis」。
3. 找到JARVIS EA v0.2資產：JARVIS Container位於「Private登錄」>「Organization Container」中。
4. 選取「JARVIS：導覽至「模型指令碼」、然後按一下「JARVIS快速入門」
5. 確認所有資產都能正常運作。
6. 尋找建立您自己應用程式的文件：PDF可在「模型指令碼」>「JARVIS文件」>「檔案瀏覽器」中找到。

"下一步：針對零售使用案例自訂狀態和流程"

針對零售使用案例自訂狀態和流程

您可以針對特定使用案例自訂對話管理程式的狀態和流程。在我們的零售範例中、我們有下列四個yaml檔案、可根據不同的意圖來引導對話。

請參閱下列每個檔案的檔案名稱和說明清單：

- 「MAIN_flow：yml」：定義主要對話流程和狀態、並在必要時將流程導向其他三個yaml檔案。
- 「REtail_flow。yml」：包含與零售或興趣點相關的狀態。系統會提供最近的零售店資訊、或是特定項目的價格。
- 「Weather（天氣）」流程：包含與天氣問題相關的狀態。如果無法判斷位置、系統會詢問後續問題以釐清。
- 「error_flow。yml」：處理使用者意圖不屬於上述三個yaml檔案的案例。顯示錯誤訊息後、系統會重新路由回接受使用者問題。下列各節包含這些yaml檔案的詳細定義。

main_flow

```
name: JarvisRetail
intent_transitions:
  jarvis_error: error
  price_check: retail_price_check
  inventory_check: retail_inventory_check
  store_location: retail_store_location
```

```
  weather.weather: weather
  weather.temperature: temperature
  weather.sunny: sunny
  weather.cloudy: cloudy
  weather.snow: snow
  weather.rainfall: rain
  weather.snow_yes_no: snowfall
  weather.rainfall_yes_no: rainfall
  weather.temperature_yes_no: tempyesno
  weather.humidity: humidity
  weather.humidity_yes_no: humidity
  navigation.startnavigationpoi: retail # Transitions should be context
and slot based. Redirecting for now.
  navigation.geteta: retail
  navigation.showdirection: retail
  navigation.showmappoi: idk_what_you_talkin_about
  nomatch.none: idk_what_you_talkin_about
states:
  init:
    type: message_text
    properties:
      text: "Hi, welcome to NARA retail and weather service. How can I
help you?"
  input_intent:
    type: input_context
    properties:
      nlp_type: jarvis
      entities:
        intent: dontcare
# This state is executed if the intent was not understood
  dont_get_the intent:
    type: message_text_random
    properties:
      responses:
        - "Sorry I didn't get that! Please come again."
        - "I beg your pardon! Say that again?"
        - "Are we talking about weather? What would you like to know?"
        - "Sorry I know only about the weather"
        - "You can ask me about the weather, the rainfall, the
temperature, I don't know much more"
      delay: 0
    transitions:
      next_state: input_intent
  idk_what_you_talkin_about:
    type: message_text_random
    properties:
```

```
      responses:
        - "Sorry I didn't get that! Please come again."
        - "I beg your pardon! Say that again?"
        - "Are we talking about retail or weather? What would you like to
know?"
        - "Sorry I know only about retail and the weather"
        - "You can ask me about retail information or the weather, the
rainfall, the temperature. I don't know much more."
      delay: 0
    transitions:
      next_state: input_intent
  error:
    type: change_context
    properties:
        update_keys:
           intent: 'error'
    transitions:
        flow: error_flow
  retail_inventory_check:
    type: change_context
    properties:
        update_keys:
           intent: 'retail_inventory_check'
    transitions:
        flow: retail_flow
  retail_price_check:
    type: change_context
    properties:
        update_keys:
           intent: 'check_item_price'
    transitions:
        flow: retail_flow
  retail_store_location:
    type: change_context
    properties:
        update_keys:
           intent: 'find_the_store'
    transitions:
        flow: retail_flow
  weather:
    type: change_context
    properties:
        update_keys:
           intent: 'weather'
    transitions:
        flow: weather_flow
```

```
  temperature:
    type: change_context
    properties:
        update_keys:
           intent: 'temperature'
    transitions:
        flow: weather_flow
  rainfall:
    type: change_context
    properties:
        update_keys:
           intent: 'rainfall'
    transitions:
        flow: weather_flow
  sunny:
    type: change_context
    properties:
        update_keys:
           intent: 'sunny'
    transitions:
        flow: weather_flow
  cloudy:
    type: change_context
    properties:
        update_keys:
           intent: 'cloudy'
    transitions:
        flow: weather_flow
  snow:
    type: change_context
    properties:
        update_keys:
           intent: 'snow'
    transitions:
        flow: weather_flow
  rain:
    type: change_context
    properties:
        update_keys:
           intent: 'rain'
    transitions:
        flow: weather_flow
  snowfall:
      type: change_context
      properties:
          update_keys:
```

```
            intent: 'snowfall'
        transitions:
            flow: weather_flow
    tempyesno:
        type: change_context
        properties:
            update_keys:
               intent: 'tempyesno'
        transitions:
            flow: weather_flow
    humidity:
        type: change_context
        properties:
            update_keys:
               intent: 'humidity'
        transitions:
            flow: weather_flow
    end_state:
      type: reset
      transitions:
        next_state: init
```

零售_流.yml

```
name: retail_flow
states:
  store_location:
    type: conditional_exists
    properties:
      key: '{{location}}'
    transitions:
      exists: retail_state
      notexists: ask_retail_location
  retail_state:
    type: Retail
    properties:
    transitions:
      next_state: output_retail
  output_retail:
      type: message_text
      properties:
        text: '{{retail_status}}'
      transitions:
        next_state: input_intent
  ask_retail_location:
```

```
    type: message_text
    properties:
      text: "For which location? I can find the closest store near you."
    transitions:
      next_state: input_retail_location
  input_retail_location:
    type: input_user
    properties:
      nlp_type: jarvis
      entities:
        slot: location
      require_match: true
    transitions:
      match: retail_state
      notmatch: check_retail_jarvis_error
  output_retail_acknowledge:
    type: message_text_random
    properties:
      responses:
        - 'ok in {{location}}'
        - 'the store in {{location}}'
        - 'I always wanted to shop in {{location}}'
      delay: 0
    transitions:
      next_state: retail_state
  output_retail_notlocation:
    type: message_text
    properties:
      text: "I did not understand the location. Can you please repeat?"
    transitions:
      next_state: input_intent
  check_rerail_jarvis_error:
    type: conditional_exists
    properties:
      key: '{{jarvis_error}}'
    transitions:
      exists: show_retail_jarvis_api_error
      notexists: output_retail_notlocation
  show_retail_jarvis_api_error:
    type: message_text
    properties:
      text: "I am having troubled understanding right now. Come again on
that?"
    transitions:
      next_state: input_intent
```

We天氣_流.yml

```yaml
name: weather_flow
states:
  check_weather_location:
    type: conditional_exists
    properties:
      key: '{{location}}'
    transitions:
      exists: weather_state
      notexists: ask_weather_location
  weather_state:
    type: Weather
    properties:
    transitions:
      next_state: output_weather
  output_weather:
      type: message_text
      properties:
        text: '{{weather_status}}'
      transitions:
        next_state: input_intent
  ask_weather_location:
    type: message_text
    properties:
      text: "For which location?"
    transitions:
      next_state: input_weather_location
  input_weather_location:
    type: input_user
    properties:
      nlp_type: jarvis
      entities:
        slot: location
      require_match: true
    transitions:
      match: weather_state
      notmatch: check_jarvis_error
  output_weather_acknowledge:
    type: message_text_random
    properties:
      responses:
        - 'ok in {{location}}'
        - 'the weather in {{location}}'
        - 'I always wanted to go in {{location}}'
      delay: 0
```

```
    transitions:
      next_state: weather_state
  output_weather_notlocation:
    type: message_text
    properties:
      text: "I did not understand the location, can you please repeat?"
    transitions:
      next_state: input_intent
  check_jarvis_error:
    type: conditional_exists
    properties:
      key: '{{jarvis_error}}'
    transitions:
      exists: show_jarvis_api_error
      notexists: output_weather_notlocation
  show_jarvis_api_error:
    type: message_text
    properties:
      text: "I am having troubled understanding right now. Come again on
that, else check jarvis services?"
    transitions:
      next_state: input_intent
```

錯誤_流.yml

```
name: error_flow
states:
  error_state:
    type: message_text_random
    properties:
      responses:
        - "Sorry I didn't get that!"
        - "Are we talking about retail or weather? What would you like to
know?"
        - "Sorry I know only about retail information or the weather"
        - "You can ask me about retail information or the weather, the
rainfall, the temperature. I don't know much more"
        - "Let's talk about retail or the weather!"
      delay: 0
    transitions:
      next_state: input_intent
```

"下一步：將第三方API連線為履約引擎"

將第三方API連線為履約引擎

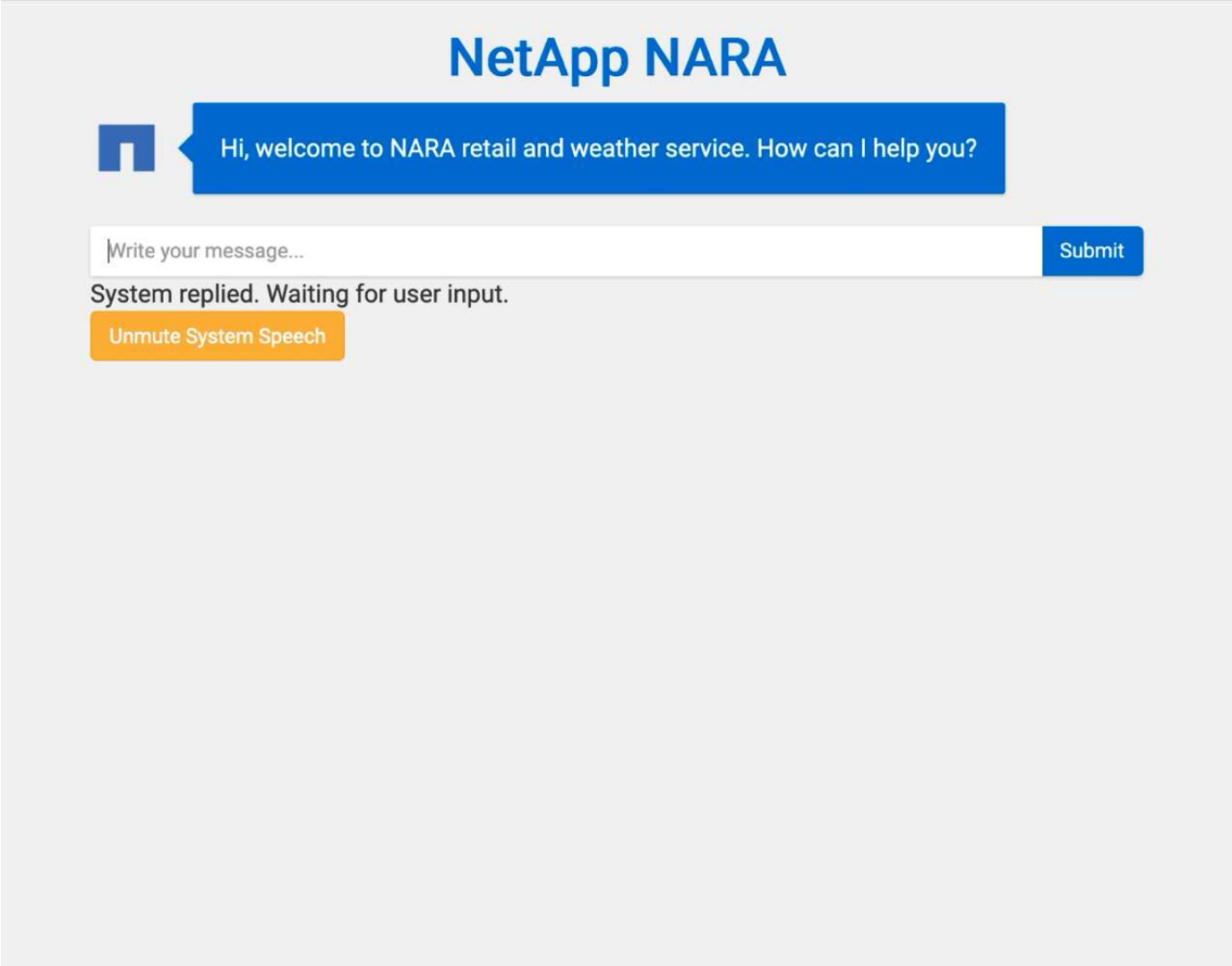
我們將下列第三方API連線為履行引擎、以回答問題：

- "WeatherStack API"：返回給定位置的天氣、溫度、雨和雪。
- "Yelp Fusion 功能API"：傳回指定位置中最近的商店資訊。
- "eBay Python SDK"：傳回指定項目的價格。

"下一步：NetApp零售助理示範"

NetApp零售助理示範

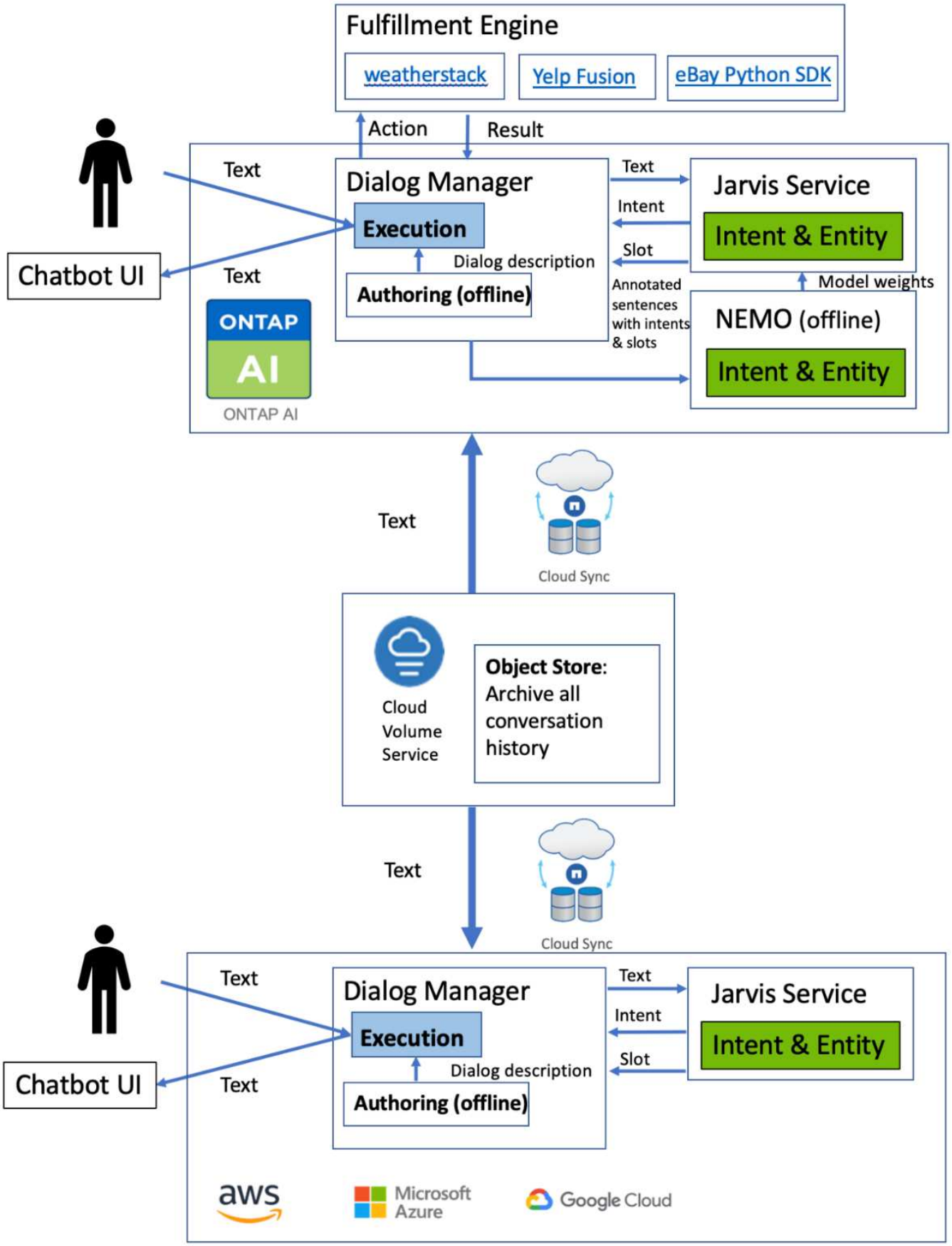
我們錄製了NetApp零售助理（Nara）的示範影片。按一下 "此連結" 開啟下圖並播放影片示範。



"下一步：使用NetApp Cloud Sync 解決方案將對話記錄歸檔"

使用NetApp Cloud Sync 解決方案將對話記錄歸檔

透過將對話歷史記錄一天轉存至CSV檔案一次、我們就能利用Cloud Sync 效益分析將記錄檔下載至本機儲存設備。下圖顯示將JARVIS部署在內部部署和公有雲中的架構、同時使用Cloud Sync 效益分析功能將對話記錄傳送給Nemo訓練。如需Nemo訓練的詳細資料、請參閱一節 "使用Nemo訓練來擴充Intent模式"。



"下一步：使用Nemo訓練來擴充Intent模式"

使用Nemo訓練來擴充Intent模式

NVIDIA Nemo是NVIDIA專為建立對話式AI應用程式所打造的工具套件。此工具套件包含預先訓練的ASR、NLP和TS模組集合、可讓研究人員和資料科學家輕鬆建立複雜的神經網路架構、並將更多焦點放在設計自己的應用程式上。

如上例所示、Nara只能處理有限類型的問題。這是因為預先訓練的NLP模式只會訓練這些類型的問題。如果我們想要讓Nara能夠處理更廣泛的問題、就必須重新訓練自己的資料集。因此、我們在此示範如何使用Nemo來延伸NLP模式、以滿足需求。首先、我們會將從Nara收集的記錄轉換成Nemo格式、然後與資料集一起訓練、以強化NLP模式。

模型

我們的目標是讓Nara根據使用者偏好來排序項目。例如、我們可能會請Nara推薦評價最高的壽司餐廳、或是想要Nara以最低價格尋找這款牛仔褲。為此、我們使用Nemo提供的意向偵測和插槽填滿模式做為訓練模式。此模式可讓Nara瞭解搜尋偏好的意圖。

資料準備

為了訓練模型、我們會收集此類問題的資料集、並將其轉換為Nemo格式。在此列出我們用來訓練模型的檔案。

dict.intents.csv

此檔案列出我們希望Nemo瞭解的所有目標。在此、我們有兩個主要目的和一個意圖、只用於將不符合任何主要目的的問題分類。

```
price_check
find_the_store
unknown
```

dict.slots.csv

此檔案列出我們訓練問題上可標示的所有插槽。

```
B-store.type
B-store.name
B-store.status
B-store.hour.start
B-store.hour.end
B-store.hour.day
B-item.type
B-item.name
B-item.color
B-item.size
B-item.quantity
B-location
B-cost.high
```

```
B-cost.average
B-cost.low
B-time.period_of_time
B-rating.high
B-rating.average
B-rating.low
B-interrogative.location
B-interrogative.manner
B-interrogative.time
B-interrogative.personal
B-interrogative
B-verb
B-article
I-store.type
I-store.name
I-store.status
I-store.hour.start
I-store.hour.end
I-store.hour.day
I-item.type
I-item.name
I-item.color
I-item.size
I-item.quantity
I-location
I-cost.high
I-cost.average
I-cost.low
I-time.period_of_time
I-rating.high
I-rating.average
I-rating.low
I-interrogative.location
I-interrogative.manner
I-interrogative.time
I-interrogative.personal
I-interrogative
I-verb
I-article
O
```

訓練：**tsv**

這是主要的訓練資料集。每一行開頭都是檔案dict.intent.csv中列出的Intent類別之後的問題。從零開始列舉標籤。

訓練插槽**.tsv**

```
20 46 24 25 6 32 6
52 52 24 6
23 52 14 40 52 25 6 32 6
…
```

訓練模型

```
docker pull nvcr.io/nvidia/nemo:v0.10
```

然後使用下列命令啟動容器。在此命令中、我們限制容器使用單一GPU（GPU ID = 1）、因為這是輕量化的訓練練習。我們也會將本機工作區/Works/nemo/對應至Container /nemo內的資料夾。

```
NV_GPU='1' docker run --runtime=nvidia -it --shm-size=16g \
                        --network=host --ulimit memlock=-1 --ulimit
stack=67108864 \
                        -v /workspace/nemo:/nemo\
                        --rm nvcr.io/nvidia/nemo:v0.10
```

在容器內、如果我們想要從原本訓練好的Bert模型開始、可以使用下列命令來開始訓練程序。data_dir是設定訓練資料路徑的引數。Work目錄可讓您設定儲存檢查點檔案的位置。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_with_bert.py \
    --data_dir /nemo/training_data\
    --work_dir /nemo/log
```

如果我們有新的訓練資料集、而且想要改善先前的模式、我們可以使用下列命令、從停止點繼續進行。Checkpoint目錄會將路徑移至先前的檢查點資料夾。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_infer.py \
    --data_dir /nemo/training_data \
    --checkpoint_dir /nemo/log/2020-05-04_18-34-20/checkpoints/ \
    --eval_file_prefix test
```

推斷模型

我們需要在經過一定次數的時間後、驗證受過訓練的模型的效能。下列命令可讓我們逐一測試查詢。舉例來說、在這個命令中、我們想要檢查我們的模式是否能正確識別查詢「哪裡可以找到最好的義大利麵」的意圖。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_infer_b1.py \
--checkpoint_dir /nemo/log/2020-05-29_23-50-58/checkpoints/ \
--query "where can i get the best pasta" \
--data_dir /nemo/training_data/ \
--num_epochs=50
```

接著、以下是推斷的輸出。在輸出中、我們可以看到我們訓練過的模型能夠正確預測future_the_store的意圖、並傳回我們感興趣的關鍵字。有了這些關鍵字、我們就能讓Nara搜尋使用者想要的內容、並進行更精確的搜尋。

```
[NeMo I 2020-05-30 00:06:54 actions:728] Evaluating batch 0 out of 1
[NeMo I 2020-05-30 00:06:55 inference_utils:34] Query: where can i get the
best pasta
[NeMo I 2020-05-30 00:06:55 inference_utils:36] Predicted intent:       1
find_the_store
[NeMo I 2020-05-30 00:06:55 inference_utils:50] where   B-
interrogative.location
[NeMo I 2020-05-30 00:06:55 inference_utils:50] can     O
[NeMo I 2020-05-30 00:06:55 inference_utils:50] i       O
[NeMo I 2020-05-30 00:06:55 inference_utils:50] get     B-verb
[NeMo I 2020-05-30 00:06:55 inference_utils:50] the     B-article
[NeMo I 2020-05-30 00:06:55 inference_utils:50] best    B-rating.high
[NeMo I 2020-05-30 00:06:55 inference_utils:50] pasta   B-item.type
```

"下一步：結論"

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